Same Direction, Different Noise

What separates rewarding right answers from penalizing wrong ones in RL for reasoning

Project brief for collaborators. Mohamed Ayman, October 18, 2026

The pitch in three sentences

When a language model is trained with reinforcement learning on math, rewarding only its correct answers and penalizing only its wrong answers push the model in **exactly the same direction** on average. The only difference between them is **noise**, and because wrong answers carry 2.5 to 3 times larger gradients, penalizing them is about 6 times noisier. This offers a measurable alternative to a well-known NeurIPS 2025 explanation that gives negative samples a special role, and our pre-registered small-scale experiments support it.

1. How we got here: the STOP idea, and why it failed

The idea. Segment-level RL splits each answer into blocks and credits every block with the change in estimated success across it. STOP asked *where the blocks should end*. Existing methods use local scores (token entropy, value variance, EPIG-Tree's gradient-aware score). We showed with exact small examples that a local score can pick the wrong cut, because a cut changes its neighbors and because token gradients interact. The plan was a geometry-aware cut rule.

The theory worked. We proved that the update error splits exactly into a part that ignores gradient geometry and an interaction part; that the interaction vanishes when token gradients are orthogonal; and that a cut's benefit depends on the other cuts only through its two neighbors.

The real model said the effect is negligible. We measured exact per-token gradients in Qwen2.5-Math-1.5B. Gradients at different positions are almost orthogonal (median $|\cos| \leq 0.005$), and the small interactions cancel in aggregate (about $\pm 1\%$ of the total). In a pre-registered test over all partitions of a 64-token grid, interactions reversed only 2.2% of orderings, none statistically clear, and never changed the best single cut. Following the geometry-free score lost at most 1.3% of the available gain.

Why it failed, in one line. The phenomenon STOP was built to exploit is real in constructed examples but too small in a real model to change decisions, so a geometry-aware method has nothing to improve on. We wrote it up honestly as "local scores suffice" (TMLR or workshop level).

What we kept. Two things that made the next project possible: exact per-answer gradient measurement tools, and the habit of measuring the mechanism before building a method.

2. WORTH: what is each generated answer worth?

The idea. Every RL-for-reasoning system decides how many answers to sample per question and how hard to push each one. These choices change the update's direction, noise, and cost at once, and they are compared by trial and error.

The theorem. For right-or-wrong rewards, the average update and the noise of *any* answer-collection rule, including adaptive ones with random group sizes, depend on only four numbers per

question: its success rate, the size of its gradient signal, and the typical gradient sizes of right and of wrong answers. This lets us compute every rule exactly and solve for the optimal one.

What the experiments said.

- Our optimal rule beat Dr. GRPO at exactly the same number of generated answers: **+1.1 and +3.3 points** on 3,382 held-out MATH problems (two seeds, paired, each statistically clear; the pre-registered bar of at least 1.5 points in both seeds was missed because of seed 1). Probe-then-fill gained +2.0 and +0.8.
- A one-line prediction (scale down wrong answers) gave no gain, and the "critical batch size" framing turned out to be already published (Bansal et al., ICML 2026 workshop).
- The decisive discovery came from measuring the four numbers: **wrong answers carry 2.4 to 3.1 times larger squared gradients than right ones**, while being only 9% longer. We found no earlier measurement; two 2026 papers assume the opposite.

3. The finding: same direction, different noise

The math. Give each right answer advantage a and each wrong answer b. The theorem gives average update $n(a-b)\nabla p$ and noise $n\,(a^2 p F_1 + b^2(1-p)F_0)$. Rewarding right answers only $(a=1, b=0)$ and penalizing wrong answers only $(a=0, b=-1)$ have the same average (classical baseline invariance; the new part is the noise). In fact, penalizing wrong answers equals rewarding right ones plus a term whose expected value is exactly zero. They differ only in noise, and with $F_0 \approx 2.5F_1$ the penalizing version is the noisier one on every question solved less than 71% of the time.

Why it matters. Zhu et al. (NeurIPS 2025) report that training only on wrong answers preserves diversity and pass@k while training only on right answers collapses it, and they credit negative samples. Our theorem says the two updates point the same way, so any difference must come from noise or implementation.

Rule (seed 1 / seed 2)	**Training acc., steps 25 to 49**	**pass@1**	**pass@64**	**Noise vs PSR**
Dr. GRPO	50.6 / 48.0	43.3 / 43.2	83.2 / 81.9	0.6×
Reward right only (PSR)	51.3 / 48.2	42.4 / 42.0	83.2 / 83.8	1×
Penalize wrong only (NSR)	33.6 / 36.4	37.7 / 39.1	83.0 / 82.1	6.4×
W-REINFORCE (Zhu et al.'s pick)	38.6 / 38.0	38.5 / 39.8	80.5 / 81.3	6×

Table 1: Qwen2.5-Math-1.5B, 150 steps, strictly on-policy. Untrained model: pass@1 13.8, pass@64 79.9.

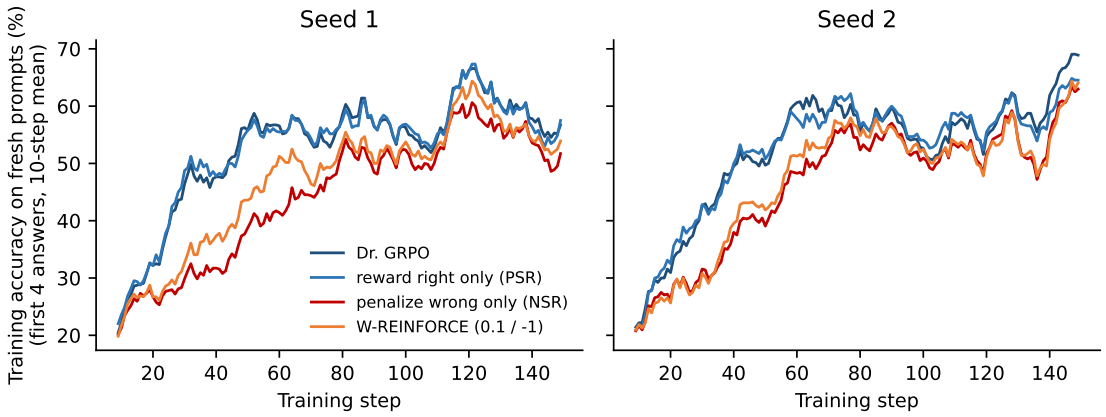


Figure 1: Training accuracy on fresh questions. The noisy rules, dominated by wrong answers, learn about twice as slowly.

Three pieces of evidence.

1. **The noise formula holds in the real model.** Measured gradient noise matches the theorem's prediction within 30% in 12 of 16 runs, including all 8 main runs; in four seed-3 baseline runs it measured 0.51 to 0.73 of the prediction, likely because gradient sizes shrink after the one-time calibration.

2. **Noise alone predicts learning speed (dose-response).** Five rules $A = r - b$ share the same average update exactly; only the noise changes. Learning speed follows the predicted noise in both seeds (rank correlation 0.9 to 0.97):

Baseline b	0.25	0 (PSR)	0.5	0.75	1 (NSR)
Predicted noise vs PSR	0.81×	1×	1.25×	2.5×	about 6×
Steps to 45% training accuracy, seed 3	37	44	44	53	56
Steps to 45% training accuracy, seed 4	38	33	41	45	56

3. **No extra diversity at equal accuracy.** Comparing pass@32 at the same pass@1, penalizing wrong answers is never above rewarding right ones beyond noise: -2.1 and -0.2 points in clean training, $+0.3$ and -1.3 points even with the NSR paper's own implementation (reused samples with clipping, per-token averaging), where at the end of training NSR's pass@32 is actually 7.0 and 1.6 points *lower* than PSR's. The intermediate baselines stay within about ± 2 points as well.

4. **Not even changing the objective moves the curve.** An update aimed exactly at pass@4 (verified to have the pass@4 gradient as its average) is only $+1.6$ and $+0.2$ points of pass@32 above Dr. GRPO at equal pass@1, and it is the same when reasoning is graded: CoT-Pass@32 is 72.7 against 72.3 (Claude Haiku 5.5 judging every correct answer's reasoning three times).

The takeaway. In the settings we tested (Qwen2.5-Math-1.5B, 150 steps), we detected no difference between the rules' accuracy-versus-diversity curves beyond the 2 to 3 points of evaluation noise (point comparisons, not yet a statistical equivalence test); the rules differ in how fast they travel along it, and the speed follows the predicted noise. This is an empirical result: the theorem

says the average raw gradients are equal, but Adam's update is not linear in the gradient, so equal averages alone would not guarantee equal training. The pre-registered "slower clock" test was formally inconclusive (two differences of 2.07 and 2.14 points against a 2-point limit), though no run showed extra diversity. Our working reading: penalizing wrong answers is a noisier, slower version of rewarding right ones, and its "preserved diversity" is what any slower run looks like. That the noise *causes* the slowdown is not yet shown; the next experiment tests it directly. W-REINFORCE, recommended by the NSR paper, was the worst rule on pass@64 in both seeds. The theory also explains why standard practice is close to right: Dr. GRPO's per-question baseline is 1.25 times less noisy than the best possible constant baseline.

4. Why this is a good paper

- **A surprising, memorable claim** that challenges a popular explanation, backed by an exact noise accounting rather than a heuristic (the equal average itself is classical).
- **A measurement we found nowhere else** (wrong answers carry 2.5 to 3 times larger gradients) that contradicts assumptions in recent variance papers.
- **Predictions made before the runs and confirmed:** the noise ordering and the learning-speed ordering (both seeds); the noise formula's level in 12 of 16 runs.
- **Rigor above the field's norm.** Every experiment was pre-registered, compared at equal numbers of generated answers, paired per problem, and run on several seeds. Our literature review found that most competing papers count training steps instead of answers and report single seeds.
- **A practical side result:** the theory's optimal answer-collection rule gains 1 to 3 points at equal cost.

5. An honest review, and what we are doing about it

We asked an independent model (Codex) to review the project as an ICML reviewer: **reject as it stands (4/10), with a credible path**; about 6/10 if the second model family shows one curve with intervals. Its main points, which we accept:

- The equal average is classical baseline invariance, and unbiased pass@k objectives exist (PKPO). What is ours is the noise accounting, the asymmetry measurement, and the matched-accuracy tests.
- "No difference detected" was presented as "the same". One 1.5B model, two seeds, 128 problems; the NSR paper used 4B to 8B models and k up to 256.
- Noise *predicts* speed, but the baseline family changes the noise and the push on wrong answers together, so cause is not shown.

Running and planned.

- **Second model family** (running): OLMo-2-1B-SFT, 300 steps, five rules, two seeds, a bootstrap equivalence test on 256 held-out problems.

- **The causal test** (designed, under review): PSR plus random-sign noise sized to match NSR's noise. Same average update, same noise, but nothing pushes wrong answers down. If it is as slow as NSR, noise is the explanation. Plus NSR at twice the learning rate and two more PSR/NSR seeds for a four-seed equivalence test.
- Venue: ICML if both hold; otherwise TMLR as a carefully scoped diagnostic paper.

Code, pre-registrations, results, and the full report: `github.com/mayman150/rlvr-adaptive` (report in `report/report.pdf`). STOP: `github.com/mayman150/STOP-RL-`.